

a great opportunity to forge a career in an unconventional yet immensely exciting industry.

Evolution of the industry

There is a significant technology shift in the industry. The current electrical architecture of a vehicle follows a distributed architecture system, meaning that multiple electronic control units (ECUs) are connected through a harness (vehicle network) and different functions of the vehicle are controlled by these ECUs.

It now follows a zone based architecture wherein multiple functions and processes inside a vehicle can be merged to function in unison on a high power computer within the vehicle.

Let's look at the BMW 7 series for instance. The luxury car model has been in India since 2015 and has roughly about 150 ECUs in it now. The ECUs support different vehicle functions but also make the vehicle a complex network system.

Although research in the field is ongoing, it is clear that the future of automotive development is going to be zone based architecture. It would support the path of automation in the vehicles as well as their electrification.

As we know, the V-model is the product development model that is widely used in the industry. It has become complex and needs adaptation with the introduction of new technologies like artificial intelligence (AI) and machine learning (ML). The development model and the development processes have therefore started incorporating agile methodologies to introduce the new technologies.

Since automation relies heavily on data, the classical V-model used in the automotive development process is slightly being adapted to have a parallel V-cycle, which is purely focusing on the data-driven software development and its testing.

Product development is not a single workflow now, it is rather becoming an iterative process with multiple parallel ongoing processes—a continuous process of development and improvement.

Where to build your career

The changing phase of how we perceive mobility and the demand of this

Can a non-engineer apply?

As you can see, the software domain is where most of the action is taking place. But software requires data. This is where having the knowledge of statistics comes into play. Even without a background in software design, it is possible to enter this domain—provided you have a good grasp on subjects like mathematics and statistics, since both help develop the analytical skills necessary to handle data. In fact, this is a trend in Europe where companies are not just hiring AI/ML engineers but looking for people with backgrounds in statistics and mathematics. So, mathematicians can do a lot in this area with the new technologies.

extremely dynamic industry have resulted in a jarring shift with regards to the skills that are required for a professional to succeed in this area. It has changed from a typical software-hardware-mechanical domain to a more detailed and more complex one. With the age of automation upon us, the software has of course become a crucial part of a vehicle's development process.

Building a career in the automotive sector is not just about building a vehicle. The complexity of ideating, developing, testing, and rolling out a product into the market has only seen an upsurge with the changing technology. There are now three broad streams comprising various job roles and skill requirements where aspiring professionals can look in to build their careers.

Development stream. Systems engineering in vehicle manufacturing is an umbrella term that incorporates the development of all key aspects like hardware, software, and mechanical. Software programming is a basic skill that you need to have, even if you are not directly associated with software development for the vehicle. It is a major skill so even fundamental knowledge in this area can go a long way.

Hardware and mechanical require different specialisations. Although they sound similar, they imply very different parts of vehicle development. The hardware deals with aspects of printed circuit board (PCB) design and development, plotting for the vehicle, standards to be followed, etc. The mechanical area deals with the mainstream design of the components and tools as well as their packaging in the vehicle. Mechanical engineering, with some background of electronics and software, plays a major role in the automation industry.

Verification and validation is another important stream to consider. It basically involves testing the vehicle

and its components and making sure that it is a quality product fit to be deployed in the market. Hence there are lots of opportunities there as well.

Supporting stream. In supporting stream the focus is on the process, methods, tools, standards, and compliance aspects. Methods and processes like Agile and ASPICE, and standards like functional safety and cyber security are pretty common in this stream.

Another common and important aspect in this stream is homologation methods. Homologation deals with the certification of a product that is to be rolled out in different geographical markets. It is an exciting field of work in with great hiring opportunities in recent times.

Tool qualification is another interesting area. Complex products that are safety-critical require a qualification process that needs to be met when testing them, and that is what tool qualification does. It is a huge market with a lot of opportunities in India, especially to provide support to different organisations outside the country.

Management stream. Finally, there is the all-important management stream. Even if becoming a manager is not your ultimate goal and you would rather handle the technical decision, technology management or the management of new product planning and introduction is a way to support your aspirations of being a technical person applying management skills.

A technical manager focuses on new product introduction brings new products and technologies into the market, along with managing the prototyping, pilot projects, and R&D. But your skills alone do not make you a project manager. You also need to gain experience and expand your knowledge of the industry. It is not something that is not very technical but executing it well is important.